Modelling Cell Proliferation in Epithelial Monolayers during *Listeria monocytogenes* infection

Lucia Orunesu[1], Raul Aparicio-Yuste[2], Maria Jose Gomez-Benito[1]

[1] *Engineering Research Institute of Aragón (I3A), University of Zaragoza, España, lorunesu@unizar.es*
[2] *Friedrich Miescher Laboratory, Tuebingen, Germany*

Introduction

Bacterial infections are a major global health burden [1]. Moreover, the increasing threat of antibiotic resistance highlights the need for new strategies to understand and combat host–pathogen interactions [2]. We focus on the intracellular pathogen *Listeria monocytogenes* (*L.m.*) and its mechanical interactions with host cells. *L.m.* enters the body through contaminated food and infects epithelial cells in the intestine. Its ability to spread from cell to cell makes it an excellent model system for studying host–pathogen interactions and intracellular bacterial infections [3]. Computational models of epithelial monolayers can help investigate how mechanical forces influence force transmission between cells, barrier function, and bacterial spread during infection. Since epithelial monolayers undergo dynamic changes in cell number and organization, we incorporate cell proliferation into our previous computational model [4]. Investigating proliferation under infected and uninfected conditions provides further insight into how bacterial infection alters the dynamics of host-cell monolayers. The model is calibrated against experimental measurements and validated using *in vitro* observations of monolayer dynamics under infected and uninfected conditions.

Methods

We use a hybrid computational model of epithelial cells that combines an agent-based model with a finite element model [4]. Cell proliferation is incorporated into the model to account for changes in cell density and monolayer dynamics over time. We account for the cell cycle by setting cells as eligible for proliferation when they are 18–30 hours old, meet an area-based proliferation criterion, and experience sufficiently high maximum principal stress, consistent with our *in vitro* observations in MDCK [5]. Infected cells are excluded from proliferation to account differences in the proliferative behaviour of healthy and infected cells. The proliferation process is integrated with the mechanical evolution of the monolayer, allowing the effects of cell division on tissue organization and collective cell dynamics to be investigated and compared with *in vitro* experiments. Model parameters are subsequently refined using Bayesian optimization [6] to match simulated cell displacements and stresses to *in vitro* measurements. The model is evaluated against experimental measurements of cell area, perimeter, displacement and mechanical stresses in infected and uninfected conditions.

Results

The computational model reproduces the mechanical behaviour observed experimentally, confirming the model's ability to capture the physiological mechanical state of the tissue (Figure 1). The increase in cell density is associated with cell proliferation, accompanied by a progressive reduction in the average cell area. The simulated temporal evolution of cell area, perimeter, cell displacement and principal stresses is compared with *in vitro* measurements. In particular, mean cell displacements decreased from approximately 1.2 to 0.6 µm in both *in silico* and *in vitro* monolayers, reproducing the experimentally observed trend. Following infection, both experiments and simulations show a similar redistribution of principal stresses, with localized stress concentrations around the infected region. Overall, the optimized simulations successfully reproduce both the stress patterns and the structural organization of the monolayer, supporting the proposed framework for describing infection-driven biomechanical changes.

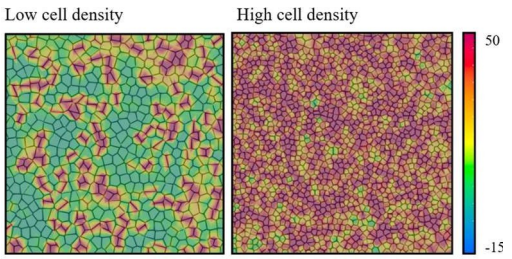


Figure 1: Temporal evolution of maximum principal stress distribution in the simulated cell monolayer during proliferation. It evolves from 625 cells (left) to approximately 1,450 cells (right).

Conclusions

The strong agreement between *in vitro* and *in silico* results validates the proposed computational model as a reliable framework for studying infection-induced biomechanical changes. This approach provides a predictive tool for investigating tissue remodelling associated with infection.

Acknowledgements

This work is part of the project PID2024-155426OB-I00 (BacHoMec), funded by MCIN/AEI/10.13039/501100011033 and by ERDF, EU.

References

[1] Ikuta KS et al. (2022). *The Lancet*, *400*(10369).
[2] Naghavi M et al. (2024). *Lancet*, 404(10459):1199–1226.
[3] Lecuit M (2020). Cellular Microbiology, 22(4)
[4] Aparicio-Yuste, R. (2025) *Comput Biol Med.*
[5] Bastounis EE et al. (2021). Dev. *Cell*, 56(4), 443–460.e11.
[6] Akiba T. et al. (2019) *Proc. ACM SIGKDD 2019*